ASSIGNMENT 1

1.Sum of two matrices.

```
#include <stdio.h>


int main()
{
   int a[10][10], b[10][10], add[10][10];
   int r1, r2, c1, c2, i, j;

   printf("Enter row and column of matrix 1: ");
   scanf("%d %d", &r1, &c1);

   printf("Enter row and column of matrix 2: ");
   scanf("%d %d", &r2, &c2);

   if(r1 != r2 || c1 != c2)
   {
      printf("Addition is not possible");
      return 0;
   }

   printf("Enter first matrix elements: ");
   for(i = 0; i < r1; i++)
   {
      for(j = 0; j < c1; j++)
      {
         scanf("%d", &a[i][j]);
      }
   }
```

```
    printf("Enter second matrix elements: ");
    for(i = 0; i < r2; i++)
    {
        for(j = 0; j < c2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            add[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("Sum of the two matrices:\n");

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            printf("%d ", add[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Output

Enter row and column of matrix 1: 2 2

Enter row and column of matrix 2: 2 2

Enter first matrix elements:

1 2

3 4

Enter second matrix elements:

5 6

7 8

Sum of the two matrices:

6 8

10 12

2.Write a C program to read n numbers into an array and display them in reverse order.

```
#include <stdio.h>

int main()
{
   int arr[50], n, i;

   printf("Enter value of n : ");
   scanf("%d", &n);

   printf("Enter the array elements:\n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);
```

```
    printf(" Reversed order: ");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", arr[i]);

    return 0;
}
```

Output

```
Enter value of n : 5
Enter the array elements:
1
2
4
5
7
 Reversed order: 7 5 4 2 1
```

3.Count the total number of duplicate elements in an array.

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, count = 0;

    printf("Enter the value of n : ");
    scanf("%d", &n);

    printf("Enter the array elements: \n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
```

```
    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d", count);

    return 0;
}
```

Output

```
Enter the value of n : 5
Enter the array elements:
6
2
4
2
4
Total number of duplicate elements = 2
```

4.Print unique elements in an array.

```
#include <stdio.h>

int main()
```

```
{
   int a[100], n, i, j, count;

   printf("Enter the value of n : ");
   scanf("%d", &n);

   printf("Enter the array elements: \n");
   for(i = 0; i < n; i++)
      scanf("%d", &a[i]);

   printf("Unique elements are: ");

   for(i = 0; i < n; i++)
   {
      count = 0;

      for(j = 0; j < n; j++)
      {
         if(a[i] == a[j])
            count++;
      }
      if(count == 1)
         printf("%d ", a[i]);
   }

   return 0;
}
```

Output

```
Enter the value of n : 6
```

Enter the array elements:

1

3

1

4

4

2

Unique elements are: 3 2

5.Insert an element in a sorted array.

```
#include <stdio.h>
int main()
{
   int a[25], n, elmt, i;
   printf("Enter value of n:");
   scanf("%d", &n);
   printf("Enter sorted elements:\n");
   for(i=0; i<n; i++)
      scanf("%d", &a[i]);

   printf("Enter element to insert: ");
   scanf("%d", &elmt);

   i = n-1;
   while(i >= 0 && a[i] > elmt)
   {
      a[i+1] = a[i];
      i--;
   }
   a[i+1] = elmt;
   n++;
```

```
    printf("Array after insertion:\n");
    for(i=0; i<n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output

```
Enter value of n: 5
Enter sorted elements:
2
4
6
8
10
Enter element to insert: 5
Array after insertion:
2 4 5 6 8 10
```

6.Insert an element in an unsorted array.

```
#include <stdio.h>
int main()
{
    int a[20], n, pos, elmt, i;
    printf("Enter value of n:");
    scanf("%d", &n);
    printf("Enter array elements:\n");
    for(i=0; i<n; i++)
        scanf("%d", &a[i]);
```

```
    printf("Enter position:");
    scanf("%d", &pos);
    if(pos > n)
    {
        printf("Invalid Position");
        return 0;
    }
    printf("Enter element to insert:");
    scanf("%d", &elmt);

    for(i=n; i>=pos; i--)
        a[i] = a[i-1];

    a[pos-1] = elmt;
    n++;

    printf("Array after insertion:\n");
    for(i=0; i<n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

Output

```
Enter value of n: 4
Enter array elements:
2
9
7
5
```

Enter position: 3

Enter element to insert: 10

Array after insertion:

2 9 10 7 5

7.Matrix multiplication.

```
#include <stdio.h>
int main()
{
   int a[10][10], b[10][10], c[10][10];
   int r1, c1, r2, c2, i, j, k;

   printf("Enter rows and columns of first matrix: ");
   scanf("%d %d", &r1, &c1);

   printf("Enter rows and columns of second matrix: ");
   scanf("%d %d", &r2, &c2);

   if(c1!= r2)
   {
      printf("Matrix multiplication not possible ");
      return 0;
   }

   printf("Enter first matrix elements:\n");
   for(i=0; i<r1; i++)
      for(j=0; j<c1; j++)
         scanf("%d", &a[i][j]);

   printf("Enter second matrix elements:\n");
   for(i=0; i<r2; i++)
```

```
        for(j=0; j<c2; j++)
            scanf("%d", &b[i][j]);

    // Initialize resultant matrix to 0
    for(i=0; i<r1; i++)
        for(j=0; j<c2; j++)
            c[i][j] = 0;

    // Matrix multiplication
    for(i=0; i<r1; i++)
    {
        for(j=0; j<c2; j++)
        {
            for(k=0; k<c1; k++)
                c[i][j] += a[i][k] * b[k][j];
        }
    }

    printf("Resultant Matrix:\n");
    for(i=0; i<r1; i++)
    {
        for(j=0; j<c2; j++)
            printf("%d ", c[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter first matrix elements:
2
4
1
3
Enter second matrix elements:
4
3
2
1
Resultant Matrix:
16 10
10  6
```

8.Matrix transpose.

```
#include <stdio.h>
int main()
{
   int a[10][10], r, c, i, j;

   printf("Enter rows and columns: ");
   scanf("%d %d", &r, &c);

   printf("Enter matrix elements:\n");
   for(i=0; i<r; i++)
      for(j=0; j<c; j++)
         scanf("%d", &a[i][j]);

   printf("Transpose matrix:\n");
```

```
    for(i=0; i<c; i++)
    {
        for(j=0; j<r; j++)
        {
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }

    return 0;
}
```

Output

```
Enter rows and columns: 3 2
Enter matrix elements:
1
2
3
4
5
6
Transpose matrix:
1 3 5
2 4 6
```

9.Right diagonal sum of a matrix.

```
#include <stdio.h>
int main()
{
    int a[10][10], n, i, j, sum = 0;
```

```
    printf("Enter order of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix elements:\n");
    for(i=0; i<n; i++)
    {
        for(j=0; j<n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i=0; i<n; i++)
        sum += a[i][n-1-i];

    printf("Right diagonal sum of matrix = %d", sum);

    return 0;
}
```

Output

```
Enter order of matrix: 2
Enter matrix elements:
2
4
6
8
Right diagonal sum of matrix = 10
```

10.Check whether a matrix is an identity matrix.

```c
#include <stdio.h>
int main()
{
    int a[10][10], n, i, j, flag = 0;

    printf("Enter order of matrix: ");
    scanf("%d", &n);

    printf("Enter matrix:\n");
    for(i=0; i<n; i++)
        for(j=0; j<n; j++)
            scanf("%d", &a[i][j]);

    for(i=0; i<n; i++)
    {
        for(j=0; j<n; j++)
        {
            if((i==j && a[i][j]!=1) || (i!=j && a[i][j]!=0))
                flag = 1;
        }
    }

    if(flag==0)
        printf("Identity Matrix");
    else
        printf("Not Identity Matrix");

    return 0;
}
```

Output

Enter order of matrix: 2

Enter matrix:

1

0

0

1

Identity Matrix

11.Find the number of vowels, consonants, digits, and white spaces in a string.

```
#include <stdio.h>
int main()
{
   char str[100];
   int i, v=0, c=0, d=0, s=0;

   printf("Enter the string: ");
   fgets(str, 100, stdin);

   for(i=0; str[i]!='\0'; i++)
   {
      if(str[i]=='a' || str[i]=='e' || str[i]=='i' || str[i]=='o' || str[i]=='u' ||
        str[i]=='A' || str[i]=='E' || str[i]=='I' || str[i]=='O' || str[i]=='U')
         v++;
      else if((str[i]>='A' && str[i]<='Z') || (str[i]>='a' && str[i]<='z'))
         c++;
      else if(str[i]>='0' && str[i]<='9')
         d++;
      else if(str[i]==' ')
         s++;
```

```
    }

    printf("Vowels = %d\n", v);
    printf("Consonants = %d\n", c);
    printf("Digits = %d\n", d);
    printf("Spaces = %d\n", s);

    return 0;
}
```

Output

```
Enter the string: Rohitha 123
Vowels = 3
Consonants = 4
Digits = 3
Spaces = 1
```

12.Find the frequency of a character in a string.

```
#include <stdio.h>
int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter the string: ");
    fgets(str, 100, stdin);

    printf("Enter the character: ");
    scanf(" %c", &ch); // note the space before %c to skip newline
```

```
    for(i=0; str[i]!='\0'; i++)
    {
        if(str[i] == ch)
            count++;
    }

    printf("Frequency = %d", count);

    return 0;
}
```

Output

```
Enter the string: rainbowcolors
Enter the character: r
Frequency = 2
```

13.Count the number of words in a string.

```
#include <stdio.h>
int main()
{
    char str[100];
    int i, words = 1;

    printf("Enter the string: ");
    fgets(str, 100, stdin);

    for(i=0; str[i]!='\0'; i++)
    {
        if(str[i] == ' ')
            words++;
    }
```

```
    printf("Words = %d", words);


    return 0;
}
```

Output

```
Enter the string: data structure homework program
Words = 4
```

14.Sort a string array.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str[10][30], temp[30];
    int n, i, j;


    printf("Enter no. of strings: ");
    scanf("%d", &n);


    printf("Enter strings:\n");
    for(i=0; i<n; i++)
        scanf("%s", str[i]);


    for(i=0; i<n-1; i++)
    {
        for(j=i+1; j<n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
```

```
            strcpy(temp, str[i]);
            strcpy(str[i], str[j]);
            strcpy(str[j], temp);
          }
       }
    }

    printf("Sorted strings:\n");
    for(i=0; i<n; i++)
       printf("%s\n", str[i]);

    return 0;
}
```

Output

```
Enter no. of strings: 5
Enter strings:
prani
rinu
kavi
varshu
Pradeepa
Sorted strings:
Kavi
Pradeepa
prani
rinu
varshu
```

15.Toggle the case of a sentence (lowercase to uppercase and vice versa).

```
#include <stdio.h>
int main()
{
   char str[100];
   int i;

   printf("Enter sentence: ");
   fgets(str, 100, stdin);

   for(i=0; str[i]!='\0'; i++)
   {
      if(str[i]>='A' && str[i]<='Z')
         str[i] = str[i] + 32;
      else if(str[i]>='a' && str[i]<='z')
         str[i] = str[i] - 32;
   }

   printf("String After Toggle:\n%s", str);

   return 0;
}
```

Output

```
Enter sentence: Anna University Guindy
String After Toggle:
aNNA uNIVERSITY gUINDY
```